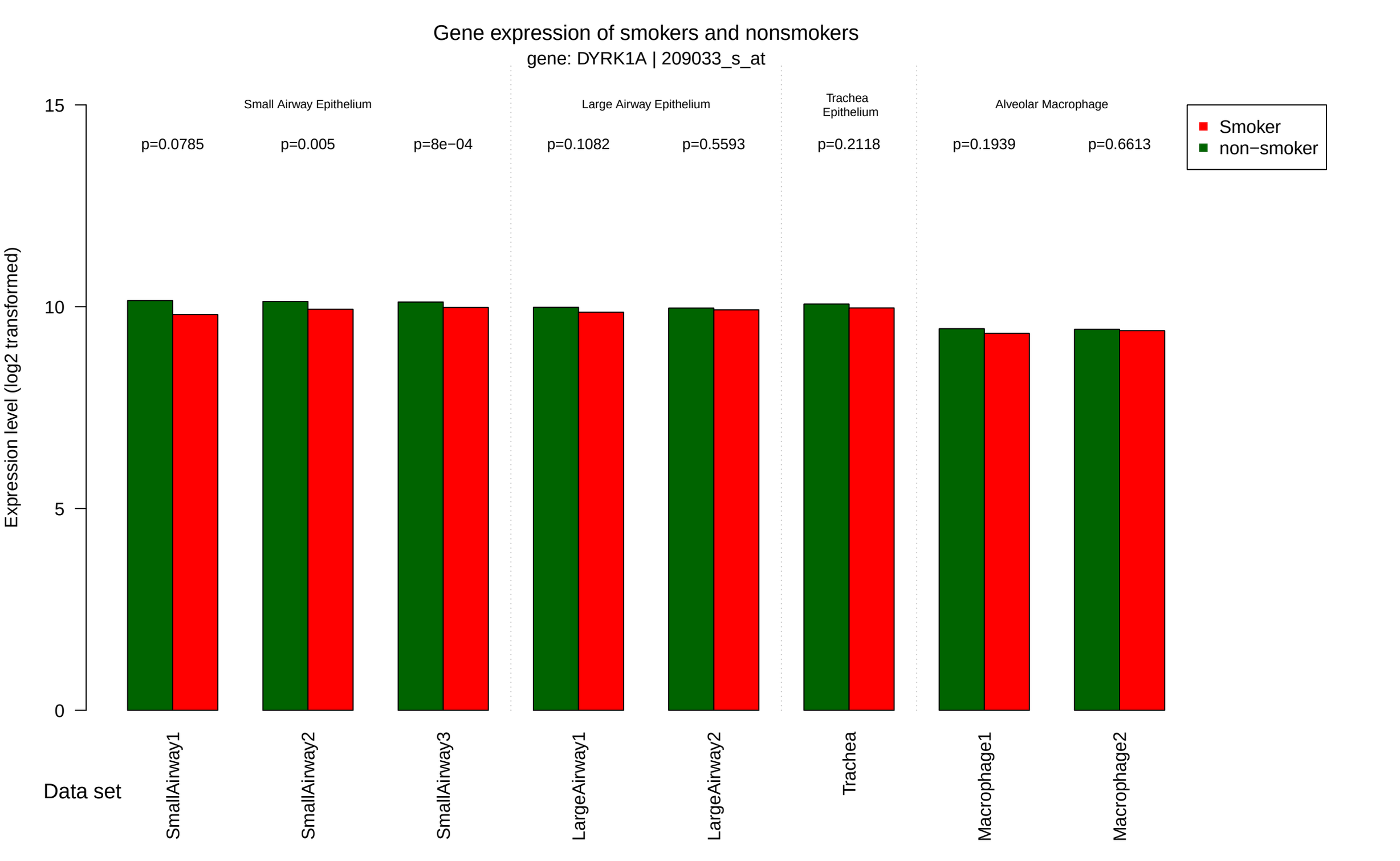
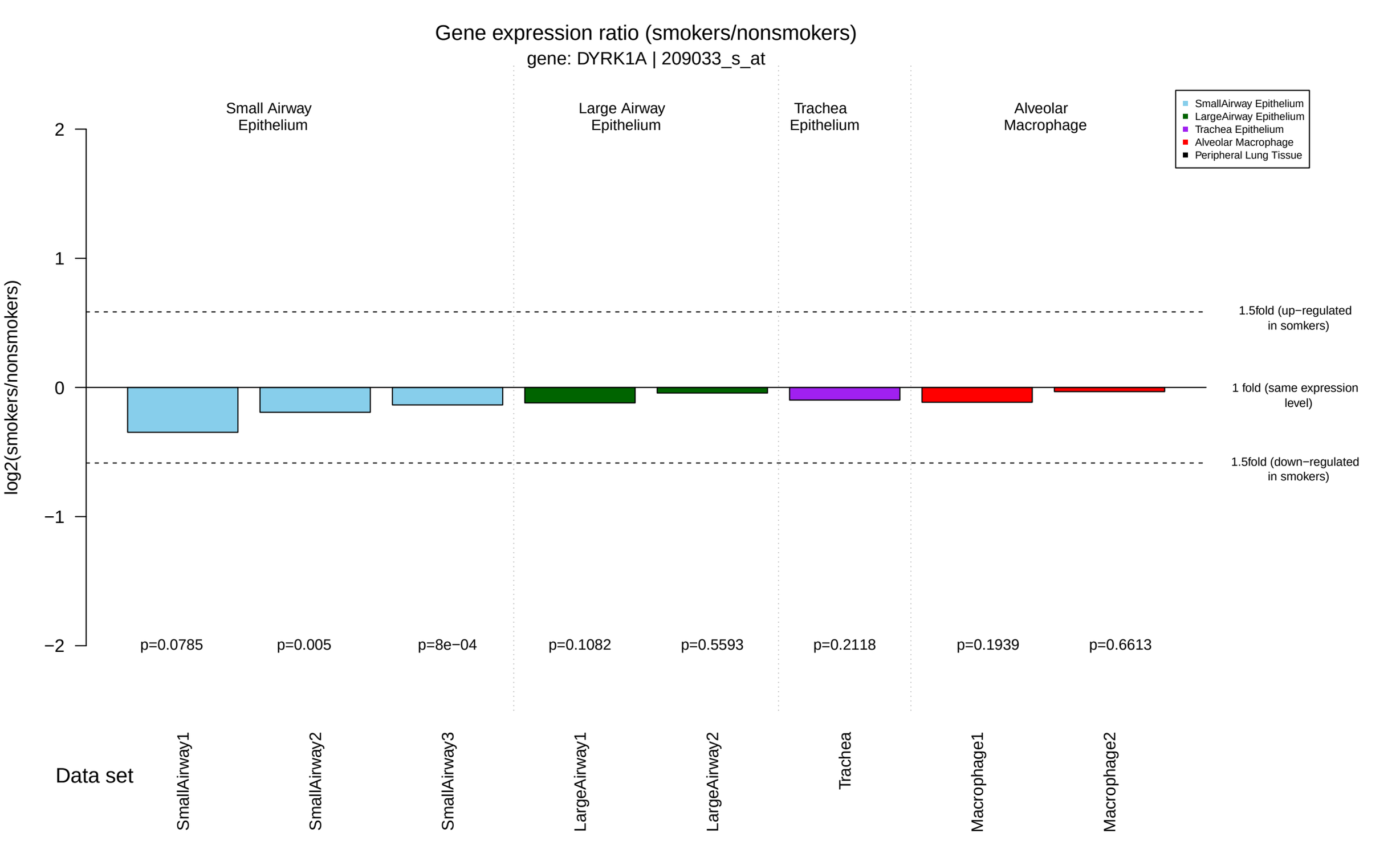
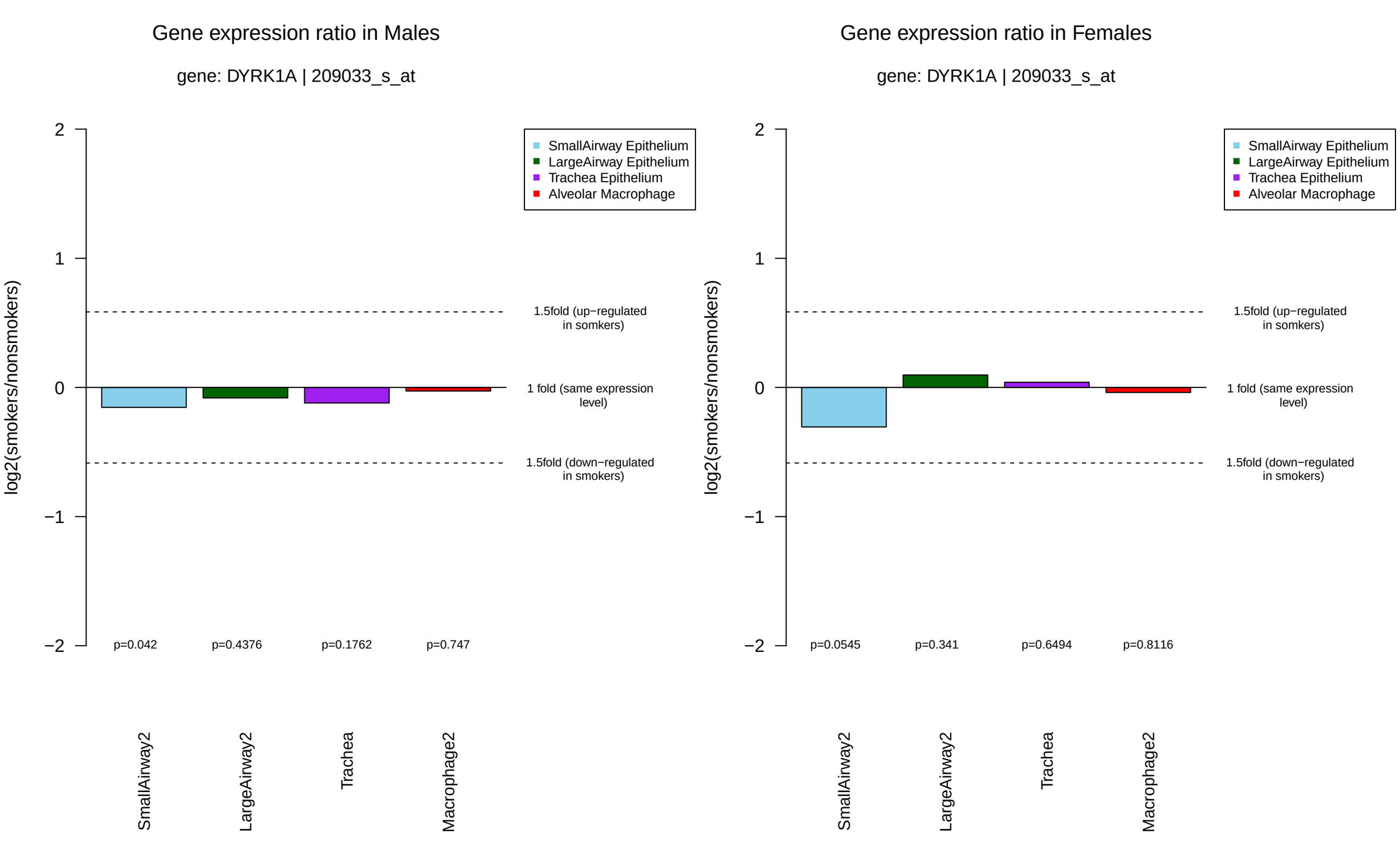
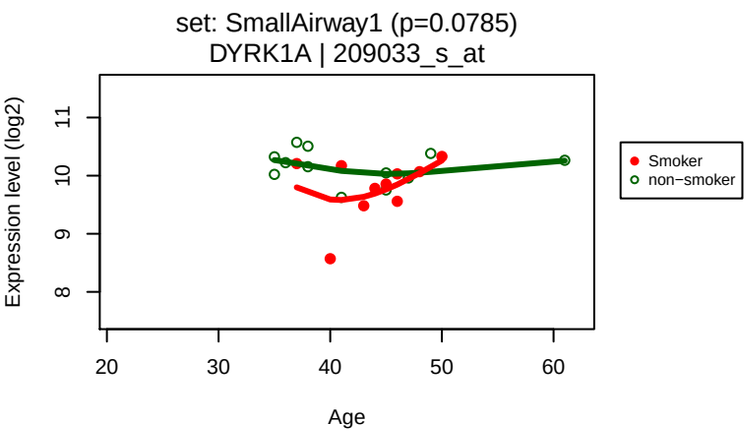
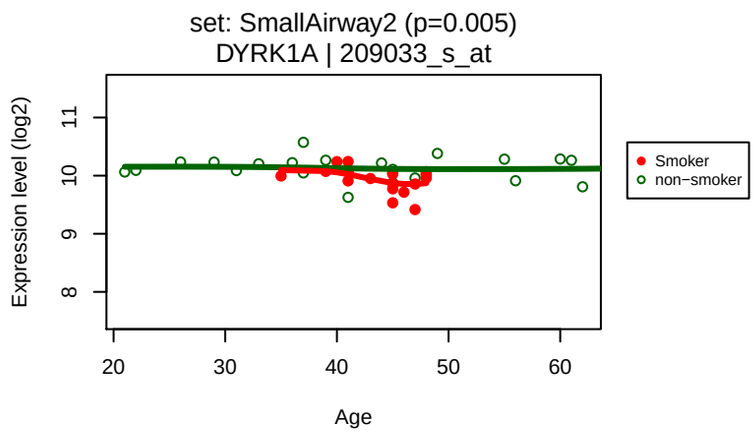
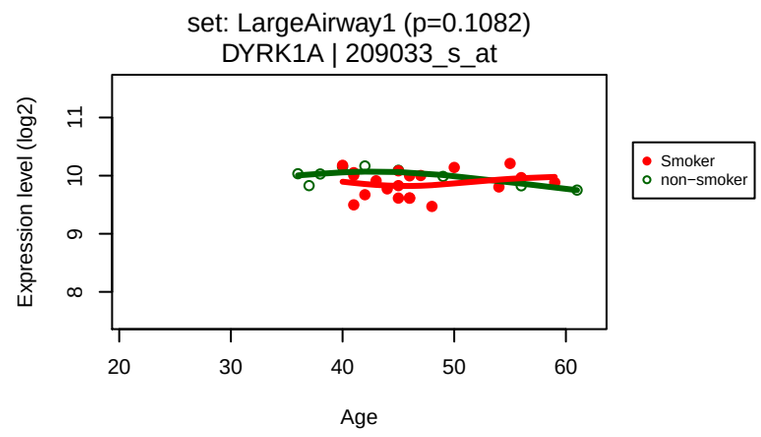

Gene expression of smokers and nonsmokers
gene: DYRK1A | 209033_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0785
p=0.005
p=8e−04
p=0.1082
p=0.5593
p=0.2118
p=0.1939
p=0.6613
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

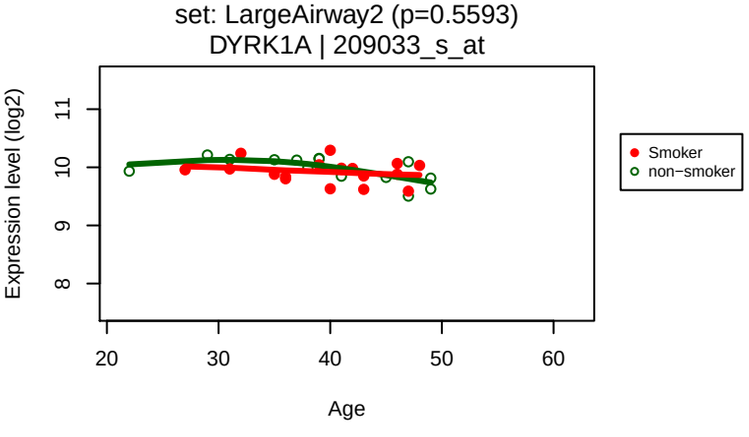

Gene expression ratio (smokers/nonsmokers)
gene: DYRK1A | 209033_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0785
p=0.005
p=8e−04
p=0.1082
p=0.5593
p=0.2118
p=0.1939
p=0.6613
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

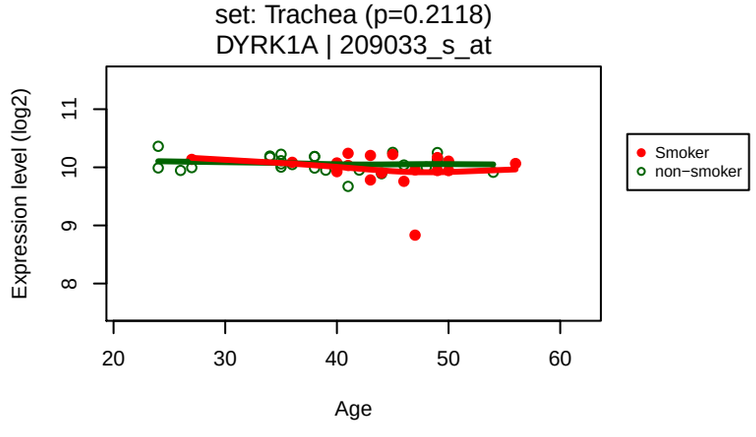

Gene expression ratio in Males
gene: DYRK1A | 209033_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.042
p=0.4376
p=0.1762
p=0.747
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: DYRK1A | 209033_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0545
p=0.341
p=0.6494
p=0.8116
SmallAirway2
LargeAirway2
Trachea
Macrophage2

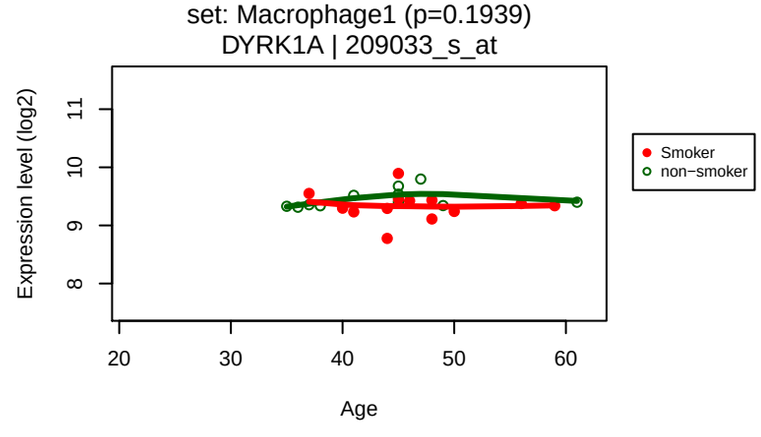

set: SmallAirway1 (p=0.0785)
DYRK1A | 209033_s_at
Expression level (log2)
Age
Smoker
non−smoker

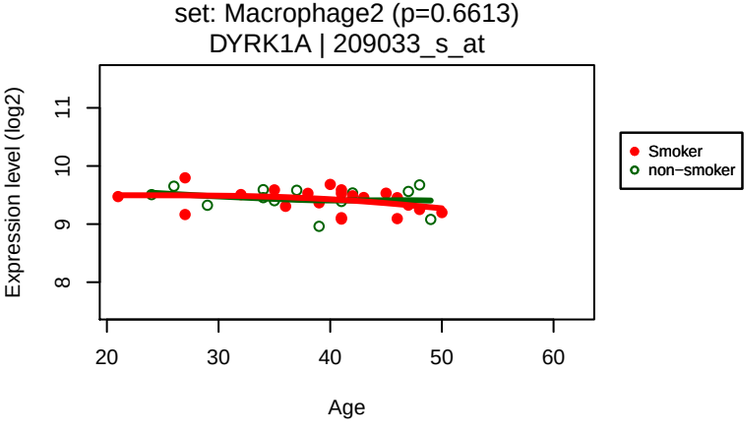

set: SmallAirway2 (p=0.005)
DYRK1A | 209033_s_at
Expression level (log2)
Age
Smoker
non−smoker

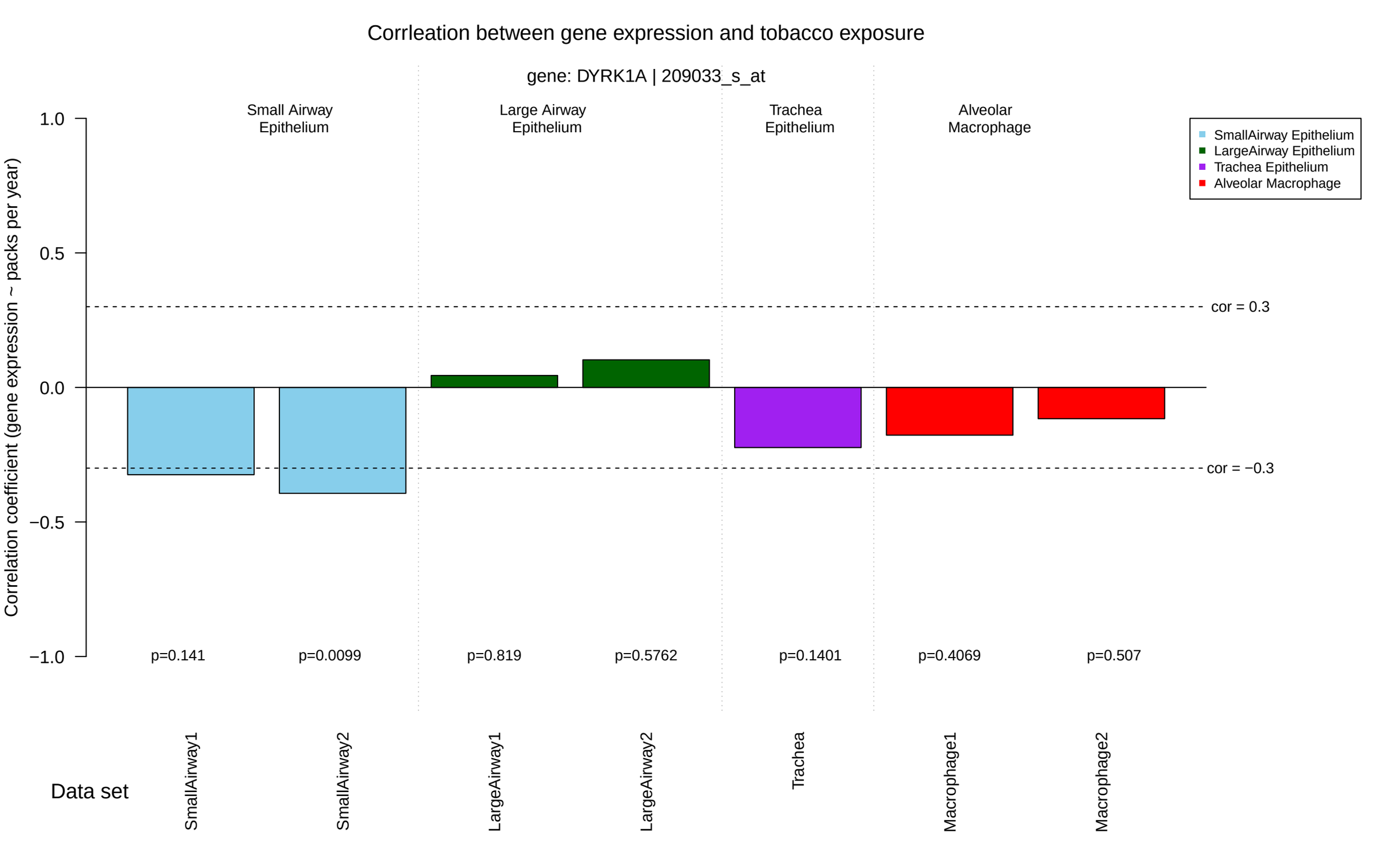

set: LargeAirway1 (p=0.1082)
DYRK1A | 209033_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5593)
DYRK1A | 209033_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2118)
DYRK1A | 209033_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1939)
DYRK1A | 209033_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6613)
DYRK1A | 209033_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: DYRK1A | 209033_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.141
p=0.0099
p=0.819
p=0.5762
p=0.1401
p=0.4069
p=0.507
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2